FAKE NEWS DETECTION USING NLP CLASSIFIERS

YADHU KRISHNAN V B
ANGEL TREESA JOHN
ANAGHA S

INTRODUCTION

- Fake news spreads rapidly through social media
- Misleads public opinion and decision-making
- Difficult to detect manually

→ **Need for automated fake news detection systems**

PROBLEM STATEMENT

Challenges in Fake News Detection

- Distinguishing fake vs real news is challenging
- Language can be intentionally misleading
- Traditional methods lack accuracy

Primary Objective

Goal: Build a robust classification system

Project Objectives

The primary goals of this research involve building a robust classification system and comparing various NLP models to effectively detect fake news.

1. **Classify news as Fake or Real**
2. **Use NLP techniques for feature extraction**
3. **Compare multiple models: Logistic Regression, Gradient Boosting, BERT**
4. **Analyze linguistic patterns**
5. **Evaluate cross-domain performance**

Dataset (LIAR)

Dataset used: LIAR Dataset

Contains: News statements with labels (Fake/Real)

Key features:

- Text content
- Metadata
- Speaker information
- Credit history

Data Preprocessing

1. **Lowercasing text**
2. **Removing punctuation**
3. **Stopword removal**
4. **Tokenization**

Goal

Converts raw text → machine-readable format

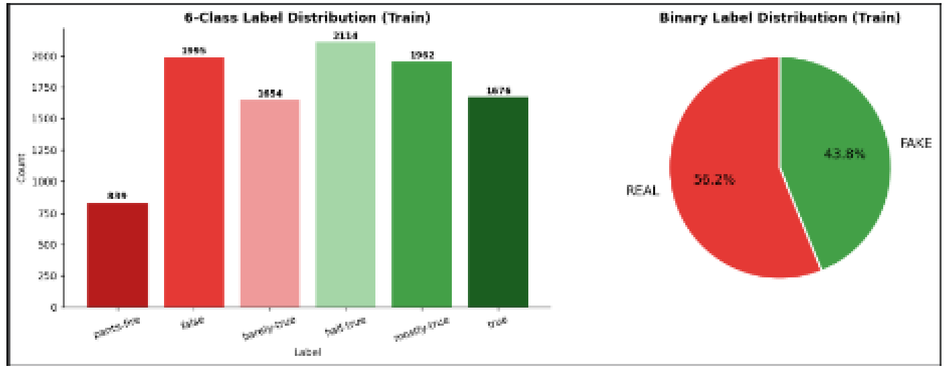
Label Distribution

The dataset originally contains 6-class labels: pants-fire, false, barely-true, half-true, mostly-true, and true.

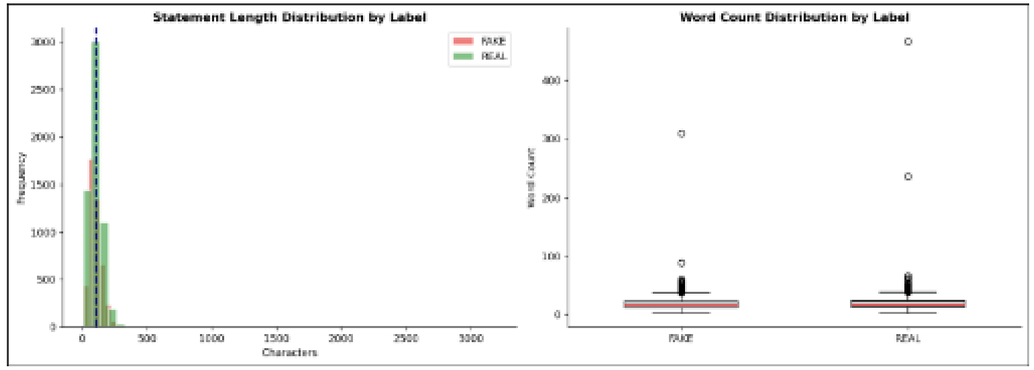
For the purpose of this study, the dataset is simplified for **binary classification**.

The binary split in the training set shows a relatively balanced distribution: **56.2% REAL** vs **43.8% FAKE**.

6-Class Label Distribution (Train)



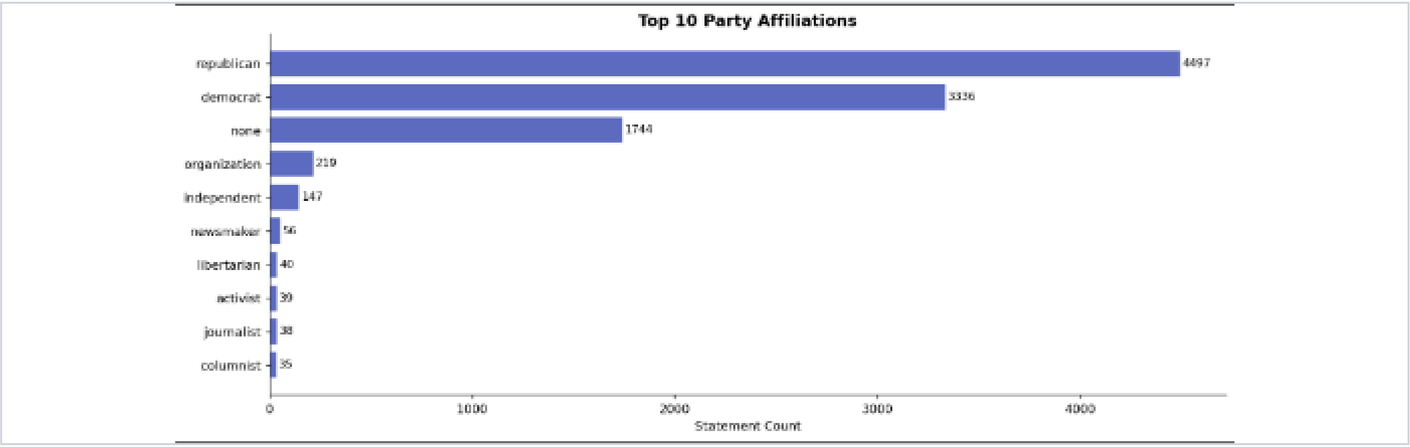
Binary Label Distribution (Train)



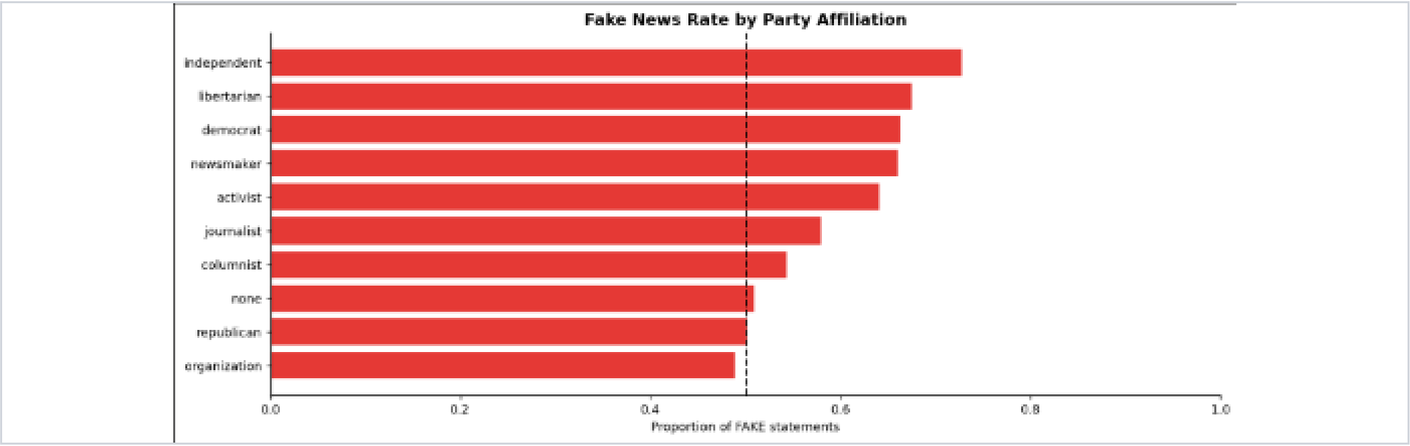
Observations

- Visual comparison of length/word-count distributions
- Fake news often uses simpler language and emotional tone (hypothesis to test)

Statement Length Distribution by Label



Word Count Distribution by Label



FEATURE ENGINEERING

Text Vectorization

TF-IDF (Term Frequency–Inverse Document Frequency)

Linguistic features

Sentence length

Readability index

Punctuation usage

Custom ratios & counts

deception_ratio

truth_ratio

number_count

Metadata

Credit history features (speaker-level)

Models Used

Traditional ML

Logistic Regression

Traditional ML baseline model using linear classification.

Traditional ML

Gradient Boosting

Ensemble technique combining weak learners for complex patterns.

Deep Learning

DistilBERT

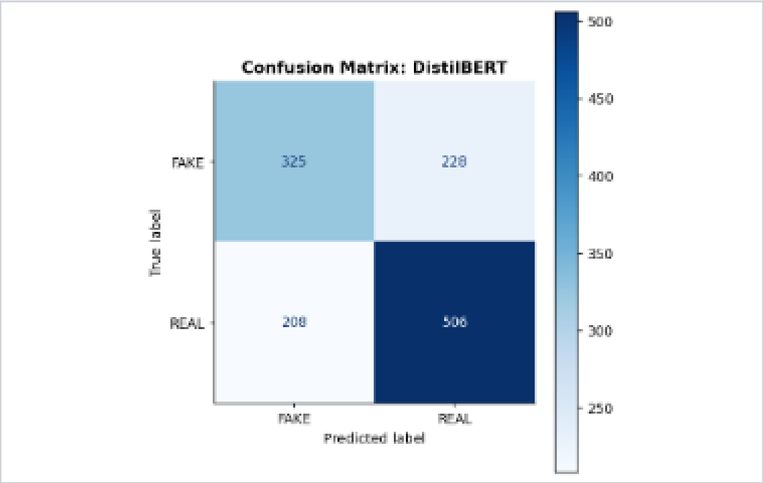
Deep Learning Transformer model for semantic understanding.

Enables comparison between simple & advanced models

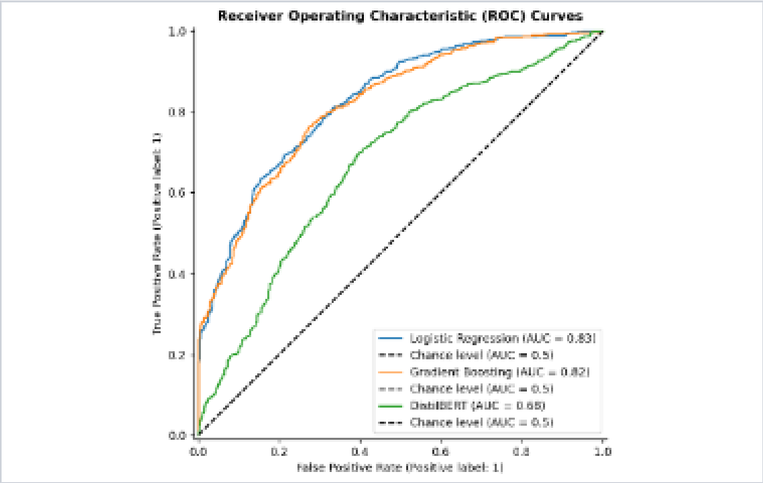
RESULTS

Model Performance Comparison

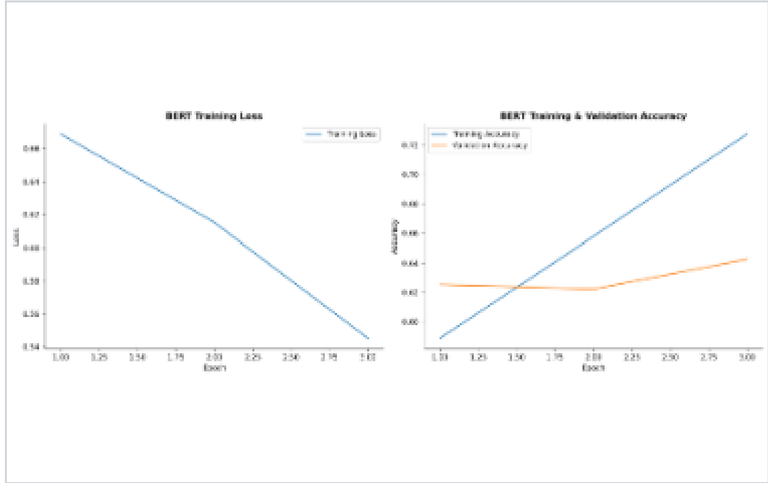
Performance Metrics



ROC Curves (AUC Comparison)



BERT Training Performance



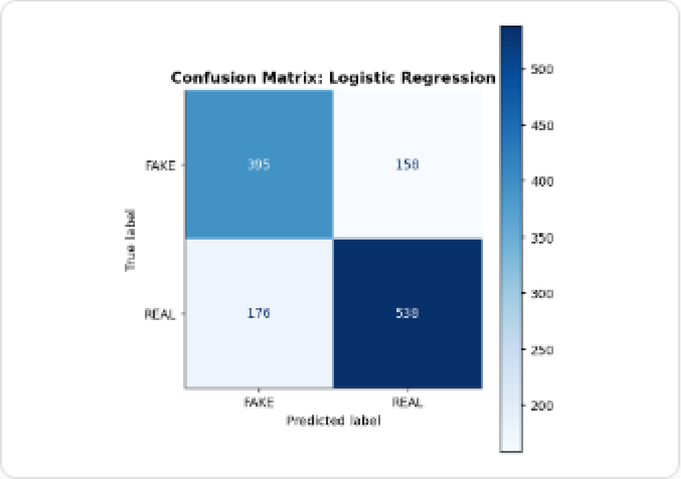
Performance Highlights

- Logistic Regression: Accuracy ~73.6%, **AUC = 0.83**
- Gradient Boosting: Accuracy ~74.3%, **AUC = 0.82**
- DistilBERT: underperforming, **AUC = 0.68**

Traditional ML outperformed the transformer model on this dataset

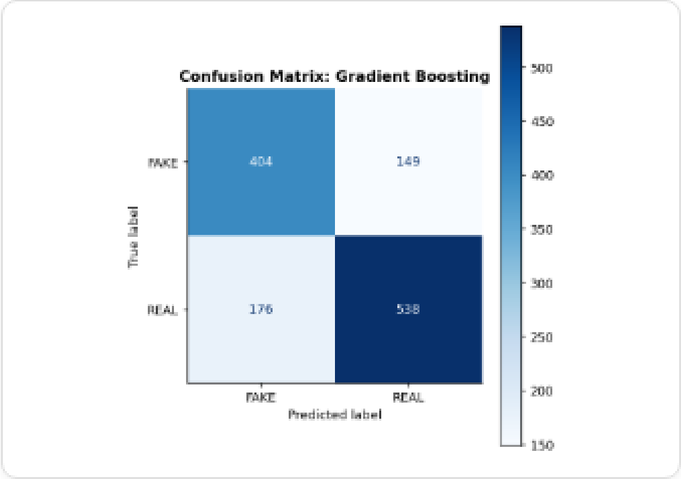
Simpler models with strong linguistic features proved more robust for this specific domain.

Confusion Matrix Analysis



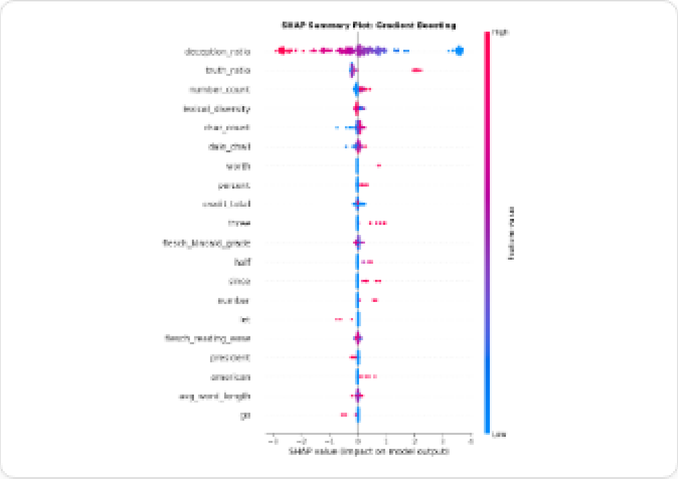
Logistic Regression

TP: 538 | TN: 395



Gradient Boosting

TP: 538 | TN: 404



DistilBERT

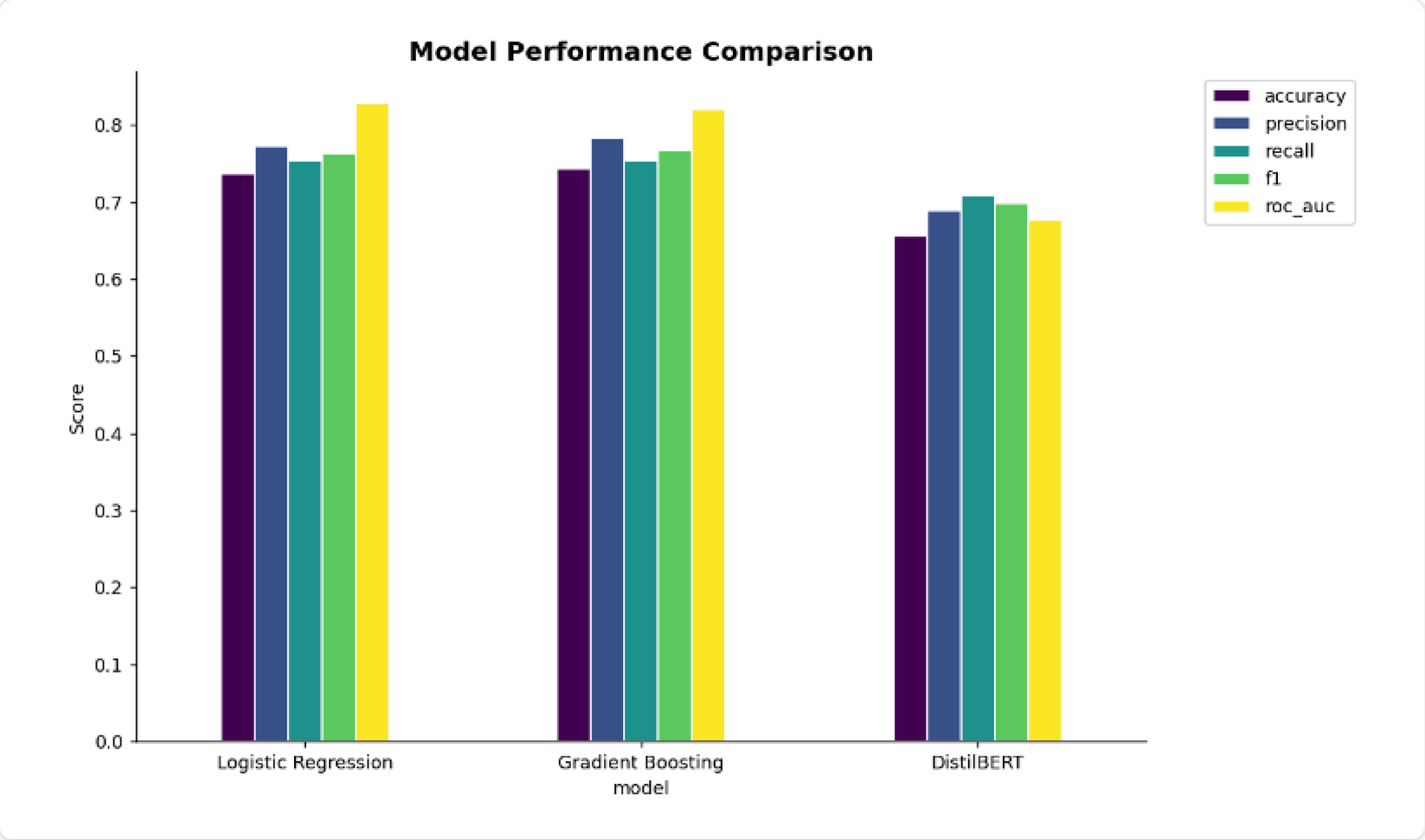
TP: 506 | TN: 325

Error Pattern Insight

Both traditional models show notable **False Negatives** (Real predicted as Fake). Gradient Boosting achieved the highest True Negative count, effectively identifying 9 more fake statements than Logistic Regression.

- Traditional ML models maintain identical performance on Real statements (TP=538).
- DistilBERT struggles significantly with True Negatives compared to ML baselines.

Model Explainability — SHAP

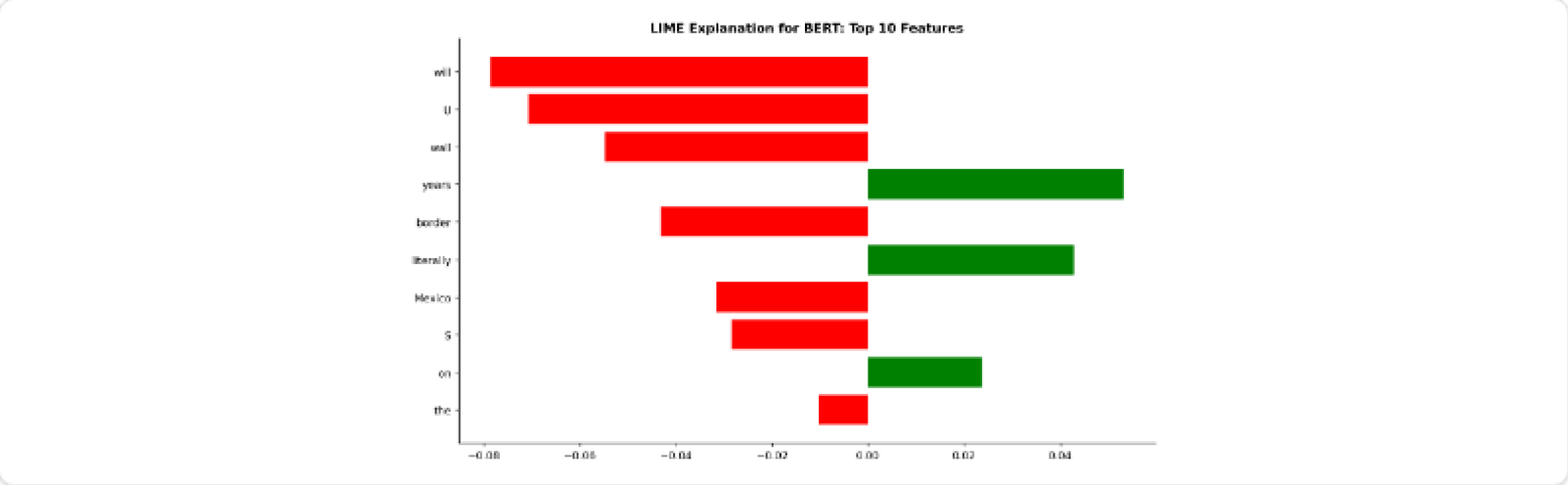


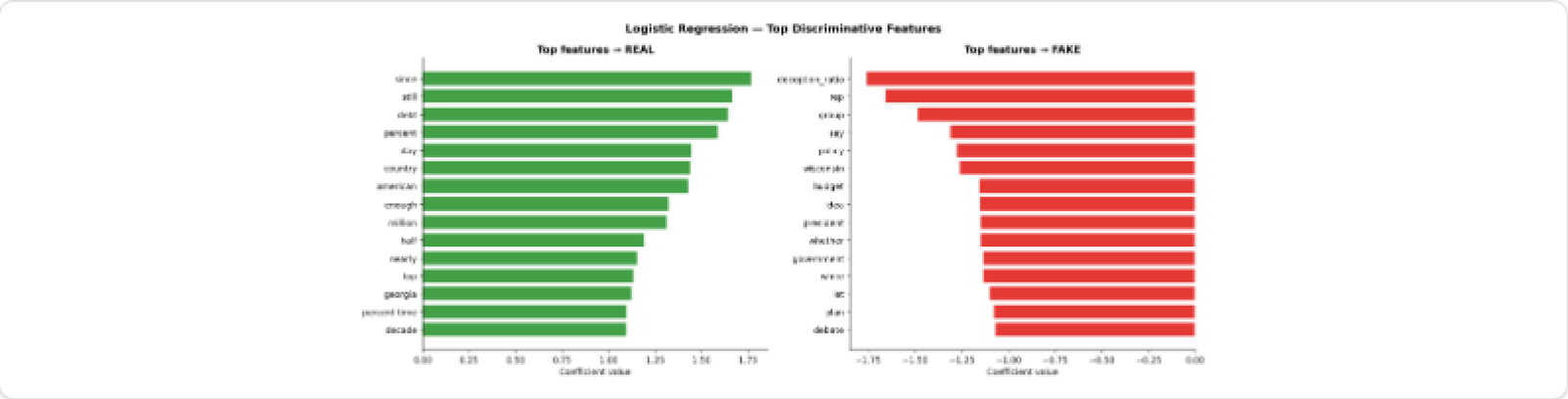
Key Insights

- Top predictor: deception_ratio
- Secondary predictors: truth_ratio, number_count

Decisions are driven by quantifiable linguistic features

Model Explainability — LIME & Feature Importance





Linguistic Signals

Influential Words (LIME)

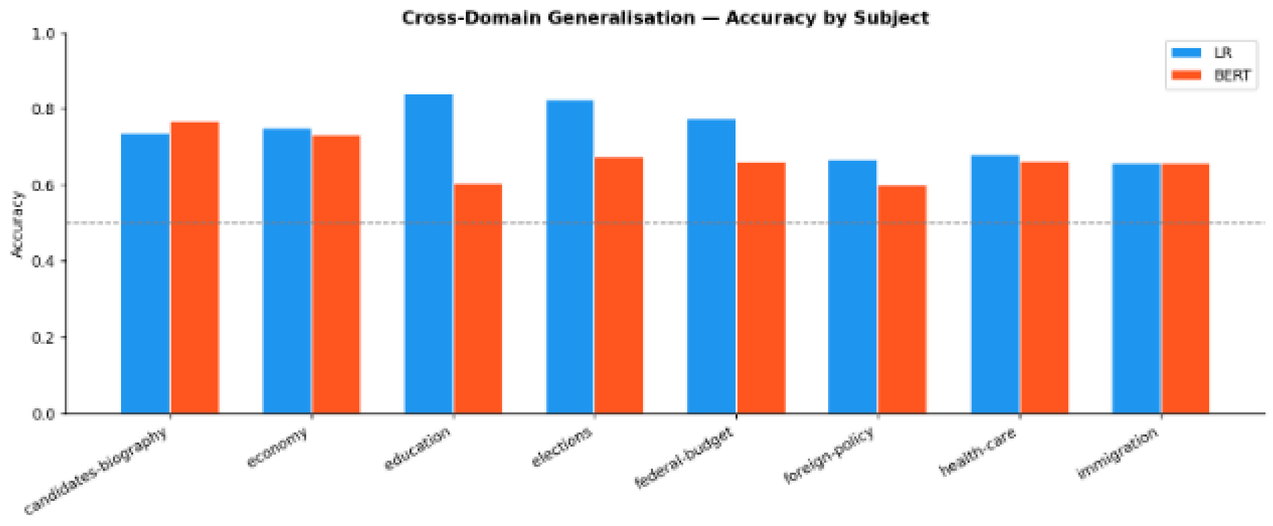
- “will”, “wall”, “Mexico”, “border”

REAL Indicators (LR)

- “since”, “still”, “debt”, “percent”

Suggests reliance on topic-specific political language

Cross-Domain Generalization



Accuracy Comparison across different news subjects

Key Observations

- LR consistently outperforms BERT across domains
- LR shows better generalization in specific subjects like foreign-policy and federal-budget
- BERT exhibits higher variance and potential overfitting to training domain

Simpler models with strong features generalize better on this dataset

Key Findings & Conclusion

- Gradient Boosting achieved best performance (small margin)
- Traditional ML outperformed DistilBERT for this dataset
- deception_ratio is the most important feature (per SHAP)
- Political/economic topics dominate the vocabulary signal
- Linguistic features improve detection

Implication: *Simpler models + strong features can beat complex models on domain-specific data. Traditional ML can be highly effective for domain-specific datasets.*

System can be used in real-world applications (with monitoring for bias)

Future Work

- Use larger datasets & Real-time detection
- Multilingual support & Address bias
- Improve transformer performance

Thank You